Earners AI and ML - by Aditya Jain sir

Lecture 08 - Recap - HW

```
What is the output?
count = 1

while count <= 5:
    count += 1

print(count)

A. 5

B. 6

C. 7

D. Error

Ans: B
```

```
count = 1

while count <= 5:
    count += 1

print(count)
```

```
6
```

```
Which situation is best suited for a while loop?

A. Print numbers from 1 to 10.

B. Print multiplication table of 5.

C. Keep asking for a password until the correct one is entered.

D. Print the first 20 even numbers.
```

```
#Keep asking for a password until the correct one is entered.

password = "Password@321"
```

```
p = ""
password = "Password@321"

while p != password:
  p = input()

print("Correct password entered!")
```

```
rt
uy
pass
Password@321
Correct password entered!
```

```
What is the output?
x = 0

while x < 3:
    print(x)
    x += 1

print("Done")
```

```
A.

0
1
2
Done

B.

1
2
3
Done

C.

0
1
2
3
Done

D. Error
```

```
x = 0

while x < 3:
    print(x)
    x += 1

print("Done")
```

```
0
1
2
Done
```

```
x = 0

while x < 3:
    print(x)
    x += 1
    print("Done")
```

```
0
Done
1
Done
2
Done
```

```
What is wrong with the following program?

count = 1

while count <= 5:
    print(count)

A. Syntax Error

B. Indentation Error

C. Infinite Loop

D. Nothing is wrong


Ans: C
```

```
# count = 1

# while count <= 5:
#     print(count)
```

```
#infinite loop
```

```
What will be printed?
number = 1

while number <= 5:
    if number % 2 == 0:
        print(number)

    number += 1

A.

1
2
3
4
5

B.

2
4

C.

1
3
5

D. Nothing
```

```
number = 1

while number <= 5:
    if number % 2 == 0:
        print(number)

    number += 1
```

```
2
4
```

```
Which of the following will create an infinite loop?

A.

count = 1

while count <= 5:
    count += 1

B.

count = 5

while count > 0:
    count -= 1

C.

count = 1

while count <= 5:
    print(count)

D.

count = 10

while count > 0:
    count -= 2
```

```
Ans: C
```

```
count = 10

while count > 0:
  print(count)
  count -= 2
```

```
10
8
6
4
2
```

```
Which statement best describes the difference between for and while loops?

A. Both loops always work the same way.

B. for loops require conditions, while while loops require range().

C. for loops are generally used when the number of iterations is known, while
while loops are used when the stopping condition is known but the number of iterations is not.

D. while loops can only work with numbers.


Ans: C
```

```
What will be printed?
count = 2

while count < 2:
    print("Hello")

print("Finished")

A.

Hello
Finished

B.

Hello
Hello
Finished

C.

Finished

D. Error


Ans: C
```

Break

```
for i in range(1,11):
  print(i)
```

```
1
2
3
4
5
6
7
8
9
10
```

```
for i in range(1,11):

  if i == 5:
    break

  print(i)
```

```
1
2
3
4
```

```
for i in range(1,11):

  print(i)

  if i == 5:
    break
```

```
1
2
3
4
5
```

```
#Keep asking for a password until the correct one is entered.

password = "Pass@321"
while True:

  p = input()

  if p == password:
    break

print("Outside the loop")
```

```
t
u
eert
rtyyjy
Pass@321
Outside the loop
```

```
for i in range(1,10,2):

  if i == 6:
    break

  print(i)
```

```
1
3
5
7
9
```

```
for i in range(0,10,2):

  if i == 6:
    break

  print(i)
```

```
0
2
4
```

Continue

```
```

```
#continue --> skip the remaining part of the current iteration, and move to the next iteration
```

```
for i in range(1,10):

  if i == 5:
    break

  print(i)
```

```
1
2
3
4
```

```
for i in range(1,10):

  if i == 5:
    continue

  print(i)
```

```
1
2
3
4
6
7
8
9
```

```
for i in range(1,10):

  print(i)

  if i == 5:
    continue
```

```
1
2
3
4
5
6
7
8
9
```

```
#Print all the even numbers between 1 to 10 (both inclusive) -- continue statement.

for i in range(1,11):

  if i%2 != 0:
    continue

  print(i)
```

```
2
4
6
8
10
```

Quiz Time - Test your learning!

```
# What will be the output?

for i in range(1, 8):

    if i == 5:
        break
```

```
    print(i)
```

```
1
2
3
4
```

```
# What will be the output?

for i in range(1, 8):

    if i == 5:
        continue

    print(i)
```

```
1
2
3
4
6
7
```

```
count = 5

while count > 0:
    print(count)
    count -= 2
```

```
5
3
1
```

```
for i in range(10):

    if i % 3 == 0:
        continue

    print(i)
```

```
1
2
4
5
7
8
```

```
for i in range(1,6):

    if i == 3:
        break

    print(i)

print("Done")
```

```
1
2
Done
```

```
num = 15

if num % 3 == 0:

    if num % 5 == 0:
        print("Both")

    else:
        print("Three")
```

```
Both
```

```
num = 10
```

```
if num % 3 == 0:

    if num % 5 == 0:
        print("Both")

    else:
        print("Three")
```

```
num = 9

if num % 3 == 0:

    if num % 5 == 0:
        print("Both")

    else:
        print("Three")
```

```
Three
```

```
for i in range(2,11,2):

    if i == 8:
        break

    print(i)
```

```
2
4
6
```

```
x = 7

while x > 0:

    if x == 4:
        break

    print(x)

    x -= 1
```

```
7
6
5
```

```
for ch in "Python":

    if ch == "h":
        continue

    print(ch)
```

```
P
y
t
o
n
```

```
i = 1

while i <= 5:

    if i == 3:
        i += 1
        continue

    print(i)

    i += 1
```

```
1
2
4
5
```

```
i = 1

while i <= 5:

    if i == 3:
        i += 1 #--> skipping this line will lead to an infinite loop
        continue

    print(i)

    i += 1
```

```
1
2
4
5
```

```
marks = 78

if marks >= 90:
    print("A")

elif marks >= 75:
    print("B")

elif marks >= 60:
    print("C")

else:
    print("D")
```

```
B
```

```
for i in range(5):

    if i == 2:
        continue

    if i == 4:
        break

    print(i)
```

```
0
1
3
```

```
for i in range(1,11):

    if i % 2 == 0:
        continue

    if i > 7:
        break

    print(i)
```

```
1
3
5
7
```

```
i = 1

while i <= 5:

    if i == 3:
        continue

    print(i)

    i += 1

#infinite loop
```

```
1
2
---------------------------------------------------------------------------
KeyboardInterrupt                         Traceback (most recent call last)
/tmp/ipykernel_1443/4093456301.py in <cell line: 0>()
      4
      5     if i == 3:
----> 6         continue
      7
      8     print(i)

KeyboardInterrupt:
```

```
number = 18

if number % 2 == 0:

    if number % 3 == 0:
        print("Divisible by both")

    else:
        print("Only Even")

else:
    print("Odd")
```

```
Divisible by both
```

```
number = 5

if number % 2 == 0:

    if number % 3 == 0:
        print("Divisible by both")

    else:
        print("Only Even")

else:
    print("Odd")
```

```
Odd
```

HW --> Rewrite bove code using else if

```
for i in range(1,6):

    if i == 2:
        continue

    print(i)

    if i == 4:
        break

A.

1 3 4

B.

1 2 3 4

C.

1 3

D.

1 2 3
```